Smart Crop Disease Detection and Yield Prediction System

B.Tech Major Project Report, 2026–27

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Abstract

Farmers lose crops to leaf diseases that are spotted late, and they plan without a quick estimate of what a field will produce. This project builds one offline web application that does both. For disease detection, each leaf photo is turned into 399 colour, texture and edge measurements and classified by a gradient boosting model into 38 crop-disease classes, trained on 11,227 PlantVillage images. For yield, a gradient boosting regressor trained on 19,689 Indian state-level records (1997–2020) estimates yield per hectare from crop, state, season, year, rainfall, fertiliser and pesticide, and scales it by field area. On held-out data the disease model reached 92.3% accuracy, and the yield model had a typical error of about 16% when predicting 2015–2020 from earlier years. The app answers "Not sure" below 50% confidence and rejects photos that are clearly not leaves. These figures come from lab-style images and state-level data, so they are upper bounds.

Keywords: crop disease detection, yield prediction, PlantVillage, gradient boosting, Flask

1. Introduction

Crop diseases reduce yield, and early detection lets a farmer act before a problem spreads. Yield estimates help with planning, storage and sale. These are usually separate problems with separate tools. This project puts both in one simple web app that runs on a single laptop without internet, so it can be shown at a farm gate or an exhibition.

Objectives

1. Identify the crop and disease from a leaf photo, with a confidence score and general crop-protection steps.
2. Estimate yield per hectare and total production for a crop, state and season.
3. Say "Not sure" instead of guessing when confidence is low.
4. Report accuracy honestly, including the limits.

Scope. In scope: 38 crop-disease classes across 14 crops, and yield for 55 crops in 30 Indian states. Out of scope: diagnosis of diseases outside these classes, soil or variety effects, and any field trial.

2. Literature Survey

Published work shows that image-based disease detection reaches very high accuracy on lab-style photos but drops sharply in the field, and that yield prediction mostly relies on weather, soil and input data.

The PlantVillage project released a large open collection of labelled leaf photos to support mobile disease diagnosis [1]. Mohanty, Hughes and Salathé trained AlexNet and GoogLeNet on these images; the best model scored 99.35% on a held-out test set, but only about 31% on online images taken under different conditions [2]. Geetharamani and Pandian used a nine-layer deep convolutional network for leaf disease identification [3]. A survey of 40 agricultural deep learning studies found that deep learning generally beats hand-crafted features with classifiers such as SVMs, but needs more data and training, and that results are hard to compare across studies [4].

For yield, a review of 50 machine learning studies found that temperature, rainfall and soil type were the most-used features, and that neural networks were the most-used algorithm [5]. Jena et al. used a functional artificial neural network on Karnataka data from 1991 to 2012 and beat a linear regression baseline [6].

Gap. High accuracy is often reported on lab images only; disease and yield are studied separately; and most systems are research code. This project adds a confidence threshold with a "Not sure" answer, puts both models behind one interface, and runs fully offline. It does not claim to beat the deep learning results above.

3. Methodology

3.1 Data

Task	Data set	Size
Leaf disease	PlantVillage colour images [1], [7] (up to 300 images per class used)	11,227 images, 14 crops, 38 classes
Yield	Indian state-level crop yield, 1997–2020 [8]	19,689 records, 55 crops, 30 states

3.2 Disease model

Each photo is resized to 128×128 and described by 399 numbers: colour histograms and statistics, fractions of brown, yellow, dark and green pixels, local binary patterns, GLCM texture statistics and HOG edge features. The data was split 70/15/15 (train/validation/test). Three classifiers (ExtraTrees, RBF-SVM, gradient boosting) were compared on validation, and the winner was scored once on the test split. A photo check rejects images with too little or too much edge detail or almost no plant colour. The app answers "Not sure" when the top probability is below 50%.

3.3 Yield model

The yield data holds state totals, so fertiliser and pesticide were converted to per-hectare rates. The model uses crop, state, season, year, rainfall, fertiliser per hectare and pesticide per hectare, and predicts log(yield) with gradient boosting. Area is not an input, because it is a state-wide total in the data; it is used only to scale production. A ridge regression baseline and a random forest were compared, and the model was tested both on a random split and on a temporal split (train to 2014, test 2015–2020).

4. Implementation

The system is a Flask server with two models behind a JSON API and a plain HTML, CSS and JavaScript interface with three tabs: Check a leaf, Estimate yield, and How well it works. It needs no internet and stores no photos. Inputs are validated (file type and size, valid categories, number ranges). Twelve automated tests cover the API, and the interface was checked at desktop and phone widths.

5. Results

5.1 Disease detection

Model	Validation accuracy
ExtraTrees (400 trees)	87.3%
SVM (RBF)	89.9%
Gradient boosting (chosen)	91.9%

On the untouched test split the chosen model reached **92.3% accuracy and 92.3% macro-F1**. On 144 unseen images sent through the live app, 91.7% were correct, with a median time of 73 ms. Raising the confidence threshold trades coverage for accuracy:

Threshold	Photos answered	Accuracy when answered
0.3	99.9%	92.4%
0.5 (used)	98.0%	93.4%
0.7	94.5%	94.7%
0.9	88.6%	96.9%

The weakest classes were tomato early blight and late blight (F1 0.79), corn grey leaf spot (0.81) and tomato Septoria leaf spot (0.83), which all produce similar brown spots. Healthy cherry, corn and strawberry leaves scored 0.99. The photo check rejected 15 of 19 non-leaf test inputs; four colourful photographs still passed.

5.2 Yield prediction

Test	Model	R^2	Median error
Random 80/20	Ridge baseline	0.766	34.2%
Random 80/20	Gradient boosting (chosen)	0.880	14.0%
Temporal 2015–2020	Gradient boosting (chosen)	0.890	16.4%
Temporal 2015–2020	Random forest	0.973	15.9%

The random forest scored higher on raw-yield R^2 but has about the same typical error, and its saved file is 67.5 MB against 2.4 MB, so gradient boosting was kept. The crop is by far the strongest input, followed by the state; rainfall, fertiliser and pesticide add little. The model mostly learns what a crop typically yields in a state.

5.3 Problems found and fixed

An early yield model used state-wide area as an input and gave impossible results for one farm; area was removed. A plain green image was once labelled with full confidence; the photo check was added. A page overflow on phones was fixed.

6. Limitations

- PlantVillage photos are lab-style; field photos will give lower accuracy [2].
- Several images can come from the same leaf, so the disease accuracy is an upper bound.

- The disease model uses hand-made features, not a deep network, because deep learning libraries could not be installed in the build environment.
- Disease coverage is 14 crops; rice and wheat are not included.
- Yield is state-level and shows typical outcomes only; it has no soil, variety or in-season weather.
- The yield file came from a public GitHub copy matching the published description [8].
- There has been no field trial. Disease advice is general; confirm with a local agriculture office.

7. Conclusion and Future Work

The project delivers a working offline web application for leaf disease detection (92.3% test accuracy over 38 classes) and yield estimation (typical error about 16%), with a "Not sure" answer, input checks and tests. Future work: replace the hand-made features with a small deep network such as MobileNet; test on real field photos with a leaf-grouped split; add rice and wheat; use district-level yield data with soil and weather inputs; and run a small field trial.

References

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